$$
\begin{aligned}
&C_{\phi\mathrm{D}}^{(11)}\ {}^{(11)} \to \frac{1}{16\,\pi^2}\\
&\Big(-\frac{1}{54}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\big[m_{\tilde{d}}{}^p\big]+\frac{5}{144}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde{d}}{}^p\big]-\frac{2}{135}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde{d}}{}^p\big]-\frac{1}{18}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\big[m_{\tilde{e}}{}^p\big]+\\
&\frac{5}{48}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde{e}}{}^p\big]-\frac{2}{45}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde{e}}{}^p\big]-\frac{1}{144}\sum_p\big(4\,g_1{}^4+9\,g_2{}^4\big)\,\mathrm{LF}_{3,0}\big[m_{\tilde{l}}{}^p\big]+\\
&\frac{1}{96}\sum_p\big(5\,g_1{}^4+6\,g_2{}^4\big)\,\mathrm{LF}_{4,-1}\big[m_{\tilde{l}}{}^p\big]-\frac{1}{45}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde{l}}{}^p\big]+\\
&\Big(\frac{3}{4}\,g_2{}^2\,\overline{y_u}^{pr}\,y_u{}^{pr}-\frac{1}{432}\sum_p\big(4\,g_1{}^4+81\,g_2{}^4\big)\Big)\,\mathrm{LF}_{3,0}\big[m_{\tilde{q}}{}^p\big]+\\
&\Big(-\frac{3}{4}\,g_2{}^2\,\overline{y_u}^{pr}\,y_u{}^{pr}+\frac{1}{288}\sum_p\big(5\,g_1{}^4+54\,g_2{}^4\big)\Big)\,\mathrm{LF}_{4,-1}\big[m_{\tilde{q}}{}^p\big]-\frac{1}{135}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde{q}}{}^p\big]-\\
&\frac{2}{27}\sum_p g_1{}^4\,\mathrm{LF}_{3,0}\big[m_{\tilde{u}}{}^p\big]+\frac{5}{36}\sum_p g_1{}^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde{u}}{}^p\big]-\frac{8}{135}\sum_p g_1{}^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde{u}}{}^p\big]-\frac{1}{36}\,g_1{}^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]-\\
&\frac{1}{24}\,g_1{}^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]+\frac{2}{45}\,g_1{}^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]-\frac{1}{32}\,g_1{}^4\,\mathrm{LF}_{2,2,-1}[m_1,\tilde{\mu}]-\frac{1}{16}\,g_1{}^4\,m_1{}^2\,\mathrm{LF}_{2,2,0}[m_1,\tilde{\mu}]-\\
&\frac{17}{32}\,g_2{}^4\,\mathrm{LF}_{2,2,-1}[m_2,\tilde{\mu}]-\frac{1}{16}\,g_2{}^4\,m_2{}^2\,\mathrm{LF}_{2,2,0}[m_2,\tilde{\mu}]+g_2{}^4\,\mathrm{LF}_{3,2,-2}[m_2,\tilde{\mu}]-\\
&\frac{3}{4}\,g_2{}^4\,m_2{}^2\,\mathrm{LF}_{3,2,-1}[m_2,\tilde{\mu}]-\frac{1}{2}\,g_2{}^4\,\mathrm{LF}_{3,3,-3}[m_2,\tilde{\mu}]+\frac{1}{2}\,g_2{}^4\,m_2{}^2\,\mathrm{LF}_{3,3,-2}[m_2,\tilde{\mu}]-\\
&\frac{1}{2}\,g_2{}^4\,\mathrm{LF}_{4,2,-3}[m_2,\tilde{\mu}]+\frac{1}{2}\,g_2{}^4\,m_2{}^2\,\mathrm{LF}_{4,2,-2}[m_2,\tilde{\mu}]-\frac{1}{6}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\mathrm{LF}_{2,2,0}\big[m_{\tilde{d}}{}^r,m_{\tilde{q}}{}^p\big]+\\
&\frac{1}{12}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde{d}}{}^r,m_{\tilde{q}}{}^p\big]-\frac{1}{6}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\mathrm{LF}_{2,2,0}\big[m_{\tilde{e}}{}^r,m_{\tilde{l}}{}^p\big]+\\
&\frac{1}{12}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde{e}}{}^r,m_{\tilde{l}}{}^p\big]+\frac{1}{6}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\mathrm{LF}_{3,1,0}\big[m_{\tilde{l}}{}^p,m_{\tilde{e}}{}^r\big]+\\
&\frac{1}{6}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde{l}}{}^p,m_{\tilde{e}}{}^r\big]-\frac{1}{8}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\big(3\,g_1{}^2+g_2{}^2\big)\,\mathrm{LF}_{4,1,-1}\big[m_{\tilde{l}}{}^p,m_{\tilde{e}}{}^r\big]+\\
&\frac{1}{6}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,y_e{}^{pr}\,\mathrm{LF}_{5,1,-2}\big[m_{\tilde{l}}{}^p,m_{\tilde{e}}{}^r\big]+\frac{1}{6}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\mathrm{LF}_{3,1,0}\big[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r\big]+\\
&\frac{1}{6}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r\big]-\frac{1}{8}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\big(5\,g_1{}^2+3\,g_2{}^2\big)\,\mathrm{LF}_{4,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r\big]+\\
&\frac{1}{2}\,g_1{}^2\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{pr}\,\mathrm{LF}_{5,1,-2}\big[m_{\tilde{q}}{}^p,m_{\tilde{d}}{}^r\big]-\frac{3}{4}\,\overline{y_u}^{pr}\,\overline{y_u}^{st}\,y_u{}^{pt}\,y_u{}^{sr}\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s\big]+\\
&\frac{3}{4}\,\overline{y_u}^{pr}\,\overline{y_u}^{st}\,y_u{}^{pt}\,y_u{}^{sr}\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s\big]+\frac{1}{12}\,g_1{}^2\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{2,2,0}\big[m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r\big]+\\
&\frac{3}{4}\,g_2{}^2\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{3,1,0}\big[m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r\big]-\frac{1}{24}\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\big(g_1{}^2+9\,g_2{}^2\big)\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r\big]-\\
&\frac{3}{4}\,g_2{}^2\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{4,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r\big]-\frac{1}{12}\,g_1{}^2\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{3,1,0}\big[m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\big]-\\
&\frac{1}{12}\,g_1{}^2\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\big]-\frac{1}{4}\,g_1{}^2\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{4,1,-1}\big[m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\big]+\\
&\frac{1}{2}\,g_1{}^2\,\overline{a_u}^{pr}\,a_u{}^{pr}\,\mathrm{LF}_{5,1,-2}\big[m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\big]-\frac{1}{16}\,g_1{}^4\,\mathrm{LF}_{3,2,-2}[\tilde{\mu},m_1]+\frac{1}{16}\,g_1{}^4\,m_1{}^2\,\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_1]+\\
&\frac{5}{24}\,g_1{}^4\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_1]+\frac{1}{16}\,g_1{}^4\,\mathrm{LF}_{4,2,-3}[\tilde{\mu},m_1]-\frac{1}{16}\,g_1{}^4\,m_1{}^2\,\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_1]-\\
&\frac{1}{6}\,g_1{}^4\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_1]+\frac{3}{16}\,g_2{}^4\,\mathrm{LF}_{3,2,-2}[\tilde{\mu},m_2]+\frac{1}{16}\,g_2{}^4\,m_2{}^2\,\mathrm{LF}_{3,2,-1}[\tilde{\mu},m_2]+\\
&\frac{5}{8}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{4,1,-2}[\tilde{\mu},m_2]+\frac{1}{16}\,g_2{}^4\,\mathrm{LF}_{4,2,-3}[\tilde{\mu},m_2]-\frac{1}{16}\,g_2{}^4\,m_2{}^2\,\mathrm{LF}_{4,2,-2}[\tilde{\mu},m_2]-\\
&\frac{1}{2}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{5,1,-3}[\tilde{\mu},m_2]+g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,2,1,-2}[m_2,\tilde{\mu},m_1]+\frac{3}{4}\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,2,1,-1}[m_2,\tilde{\mu},m_1]-\\
&\frac{1}{2}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,2,1,-3}[m_2,\tilde{\mu},m_1]-\frac{1}{2}\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,2,1,-2}[m_2,\tilde{\mu},m_1]+\\
&\frac{3}{4}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,y_d{}^{sr}\,\overline{y_u}^{st}\,y_u{}^{pt}\,\mathrm{LF}_{2,2,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s,m_{\tilde{d}}{}^r\big]-\frac{3}{4}\,\overline{y_u}^{st}\,y_u{}^{pt}\,\overline{a_u}^{pr}\,a_u{}^{sr}\,\mathrm{LF}_{2,1,1,0}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s,m_{\tilde{u}}{}^r\big]+\\
&\frac{3}{4}\,\overline{y_u}^{st}\,y_u{}^{pt}\,\overline{a_u}^{pr}\,a_u{}^{sr}\,\mathrm{LF}_{3,1,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s,m_{\tilde{u}}{}^r\big]+\frac{3}{8}\,\overline{y_u}^{st}\,y_u{}^{pt}\,\overline{a_u}^{pr}\,a_u{}^{sr}\,\mathrm{LF}_{2,2,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^s\big]-\\
&\frac{3}{4}\,\overline{y_u}^{st}\,y_u{}^{pt}\,\overline{a_u}^{pr}\,a_u{}^{sr}\,\mathrm{LF}_{2,1,1,0}\big[m_{\tilde{q}}{}^s,m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r\big]+\frac{3}{4}\,\overline{y_u}^{st}\,y_u{}^{pt}\,\overline{a_u}^{pr}\,a_u{}^{sr}\,\mathrm{LF}_{3,1,1,-1}\big[m_{\tilde{q}}{}^s,m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r\big]+\\
&\frac{3}{8}\,\overline{y_u}^{st}\,y_u{}^{pt}\,\overline{a_u}^{pr}\,a_u{}^{sr}\,\mathrm{LF}_{2,2,1,-1}\big[m_{\tilde{q}}{}^s,m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^p\big]-\frac{7}{16}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},m_1,m_2]-\\
&\frac{1}{8}\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{2,1,1,0}[\tilde{\mu},m_1,m_2]+\frac{3}{8}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,1,1,-2}[\tilde{\mu},m_1,m_2]+\\
&\frac{1}{8}\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,1,1,-1}[\tilde{\mu},m_1,m_2]-\frac{1}{8}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{4,1,1,-3}[\tilde{\mu},m_1,m_2]-\\
&\frac{1}{8}\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{4,1,1,-2}[\tilde{\mu},m_1,m_2]-\frac{1}{2}\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,2,1,-3}[\tilde{\mu},m_2,m_1]-\\
&\frac{1}{2}\,m_1\,m_2\,g_1{}^2\,g_2{}^2\,\mathrm{LF}_{3,2,1,-2}[\tilde{\mu},m_2,m_1]-\frac{1}{8}\,\tilde{\mu}^4\,\overline{y_e}^{pr}\,\overline{y_e}^{st}\,y_e{}^{pt}\,y_e{}^{sr}\,\mathrm{LF}_{2,2,1,1,-1}\big[m_{\tilde{l}}{}^p,m_{\tilde{l}}{}^s,m_{\tilde{e}}{}^r,m_{\tilde{e}}{}^t\big]-\\
&\frac{3}{8}\,\tilde{\mu}^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st}\,y_d{}^{pt}\,y_d{}^{sr}\,\mathrm{LF}_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s,m_{\tilde{d}}{}^r,m_{\tilde{d}}{}^t\big]+\frac{3}{4}\,\tilde{\mu}^2\,\overline{y_d}^{st}\,y_d{}^{pt}\,\overline{a_u}^{pr}\,a_u{}^{sr}\\
&\quad\mathrm{LF}_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s,m_{\tilde{d}}{}^t,m_{\tilde{u}}{}^r\big]-\frac{3}{4}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,a_u{}^{pt}\,a_u{}^{sr}\,\mathrm{LF}_{2,1,1,1,0}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s,m_{\tilde{u}}{}^r,m_{\tilde{u}}{}^t\big]+\\
&\frac{3}{4}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,a_u{}^{pt}\,a_u{}^{sr}\,\mathrm{LF}_{3,1,1,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{q}}{}^s,m_{\tilde{u}}{}^r,m_{\tilde{u}}{}^t\big]+\frac{3}{8}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,a_u{}^{pt}\,a_u{}^{sr}\\
&\quad\mathrm{LF}_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^r,m_{\tilde{q}}{}^s,m_{\tilde{u}}{}^t\big]+\frac{3}{8}\,\overline{a_u}^{pr}\,\overline{a_u}^{st}\,a_u{}^{pt}\,a_u{}^{sr}\,\mathrm{LF}_{2,2,1,1,-1}\big[m_{\tilde{q}}{}^p,m_{\tilde{u}}{}^t,m_{\tilde{q}}{}^s,m_{\tilde{u}}{}^r\big]\Big)
\end{aligned}
$$